

Identity Services Lab 1

Local User Authentication and Secure SSH

Cisco Packet Tracer Evidence Portfolio

Project Objective

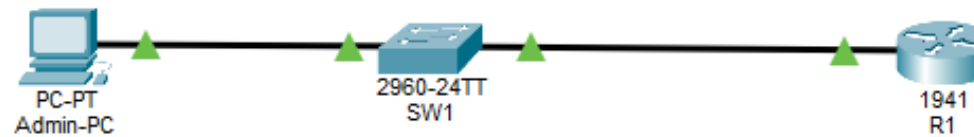
Build and secure a small Cisco network by configuring local administrator identities, encrypted credentials, SSH version 2, console and VTY authentication, management addressing, security banners, verification tests, and saved startup configurations.

Skills Demonstrated

Identity and access management | Cisco IOS | Secure Shell (SSH) | IPv4 networking | Local authentication | Network troubleshooting | Configuration verification | Secure device administration

1. Network Topology

al x: 221, y: 2



Cisco Packet Tracer topology containing an administrator workstation, a Cisco 2960 switch, and a Cisco 1941 router with all links operational.

[Topology design](#) | [Device naming](#) | [Cabling](#) | [Interface status verification](#)

2. Connectivity Test to R1

```
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Successful ICMP replies confirm that Admin-PC can reach the router at 192.168.10.1 with zero packet loss.

[IPv4 addressing](#) | [Default gateway configuration](#) | [Ping testing](#) | [Troubleshooting](#)

3. Successful Local User SSH Authentication

```
C:\>ssh -l admin 192.168.10.1
```

```
Password:
```

```
AUTHORIZED USERS ONLY
```

```
R1#
```

Admin-PC connects securely to R1 through SSH using a locally configured administrator account and receives the authorized-use banner.

[Local identities](#) | [SSH version 2](#) | [Privilege level 15](#) | [Secure remote administration](#)

4. Active SSH Session and Version Verification

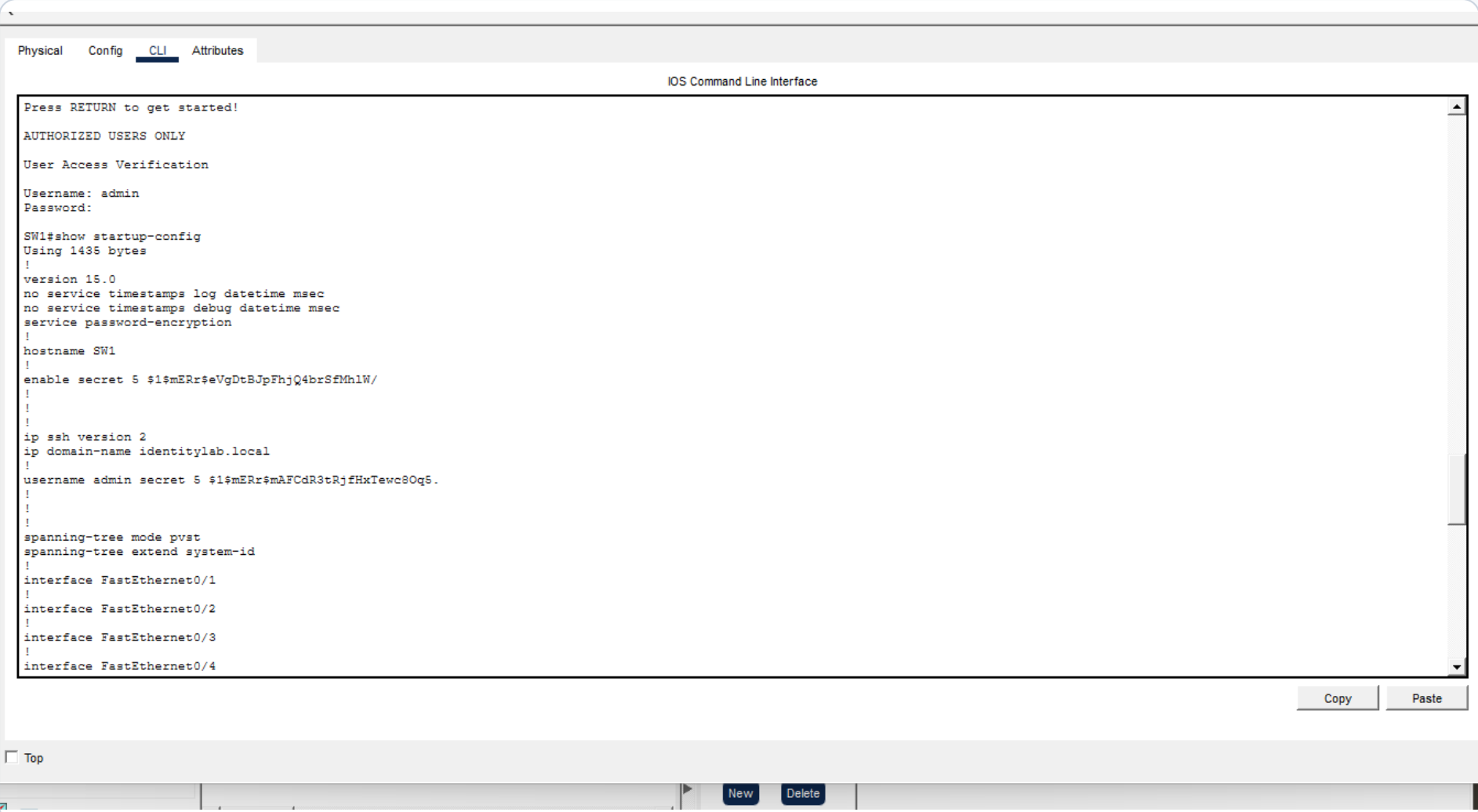
```
R1#show users
      Line      User      Host(s)      Idle      Location
*134 vty 0      admin      idle         00:00:00

      Interface  User      Mode      Idle      Peer Address
R1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
R1#
```

The show users and show ip ssh commands verify the active administrator session and confirm that SSH version 2 is enabled.

[Session verification](#) | [Cisco IOS show commands](#) | [Authentication validation](#) | [SSH hardening](#)

5. Saved SW1 Identity and SSH Configuration



The screenshot shows the Cisco Packet Tracer interface with the CLI tab selected. The window title is "IOS Command Line Interface". The text inside the CLI window is as follows:

```
Press RETURN to get started!  
AUTHORIZED USERS ONLY  
User Access Verification  
Username: admin  
Password:  
SW1#show startup-config  
Using 1435 bytes  
!  
version 15.0  
no service timestamps log datetime msec  
no service timestamps debug datetime msec  
service password-encryption  
!  
hostname SW1  
!  
enable secret 5 $1$mERr$eVgDtBJpFhjQ4brSfMh1W/  
!  
!  
!  
ip ssh version 2  
ip domain-name identitylab.local  
!  
username admin secret 5 $1$mERr$mAFcdR3tRjfHxTewc8Oq5.  
!  
!  
!  
spanning-tree mode pvst  
spanning-tree extend system-id  
!  
interface FastEthernet0/1  
!  
interface FastEthernet0/2  
!  
interface FastEthernet0/3  
!  
interface FastEthernet0/4
```

At the bottom of the CLI window, there are "Copy" and "Paste" buttons. Below the CLI window, there is a "Top" button and a "New" button.

The startup configuration confirms the switch hostname, encrypted enable secret, local administrator account, domain name, and SSH version 2 settings.

[Configuration persistence](#) | [Encrypted credentials](#) | [Switch management](#) | [Identity services](#)